

Assignment 1

Faculty: MZG

1. You are given a sorted array of 8000 unique integers and you need to search for the target number 4561. [2.5+2.5]
 - a. What is the maximum number of comparisons needed in binary search in the worst case?
 - b. What is the maximum number of comparisons in quaternary search (i.e., splitting into 4 parts)?
2. Suppose you are working in a resource-constrained environment where memory is very limited, and you must sort a huge dataset. You have the choice between QuickSort and Merge Sort. Which one should you choose, and why? Consider both time and space complexities in your answer. [2.5]
3. You have decided to implement Quick Sort using the first element of the array as the pivot to sort a dataset. Will this pivot selection approach be efficient for an already sorted or nearly sorted array? Explain your reasoning in detail. [2.5]
4. Sort the following functions in ascending order: [5]
 $f_1 = n^2 \log n$, $f_2 = n^{4/5}$, $f_3 = n^{\log n}$
5. You have an array where the left subarray is in strictly decreasing order and the right subarray is in strictly increasing order. Propose a modified version of your ternary search algorithm. [5]

Algorithms: Assignment 3

Faculty : MZG

1. Solve the recurrent relations using either Recursion Tree or Back Substitution approach:
 - a. $T(n) = 4T(n/2) + n$ [3]
 - b. $T(n) = 2T(n/2) + n \log n$ [3]
2. Use any of the three approaches to solve the following recurrent relations:
 - a. $T(n) = 2T(n/2) + n/\log n$ [2]
 - b. $T(n) = T(\sqrt{n}) + 1$ [2]
 - c. $T(n) = T(n/2) + T(n/4) + n$ [2]
3. State which Sorting algorithm you will use on the scenarios below and explain (in 2 sentences MAX.) the reasoning behind your choice. Do not assume additional contexts in these scenarios.
 - a. [0.5, 0.1, 0.2, 0.4, 10] must be sorted in ascending order with very limited memory. [2]
 - b. The data is in random order but ensuring good performance is critical. [2]
 - c. The elements are [10000, 2, 4] and must be sorted in ascending order while ensuring the best system performance with limited memory. [2]
 - d. You repeatedly sort very tiny arrays inside a larger algorithm. [2]

CSE221: Algorithms

Assignment 3, MZG

1. In the DFS based cycle detection problem for undirected graphs, how will you modify the code to count the number of cycles. Write a pseudocode for your approach. [5]
2. Why does a back edge imply a cycle? Can a back edge go to a parent for directed and undirected graphs? [2.5+2.5]
3. You are given a directed graph. Consider the following modified pseudocode:

```
Kosaraju(G):  
    stack = empty  
    visited = [false]*n  
  
    for each vertex u:  
        if not visited[u]:  
            DFS1(u)  
  
    transpose G -> GT  
  
    visited = [false]*n  
    while stack not empty:  
        u = stack.pop()  
        if not visited[u]:  
            DFS2(u)  
            print "SCC found"
```

- Now suppose DFS1 pushes nodes to stack before exploring neighbors instead of after. Will SCCs still be correctly identified and if not, explain in detail why not? [5]
4. Dijkstra is run on a graph with one negative edge but no negative cycles. Will it always Fail? Explain using simulations. [2.5]
 5. Are maximum weighted edges always excluded from Minimum Spanning Trees? [2.5]

Assignment 4

CSE221: Algorithms

- 1a. Briefly explain the time and space complexity of Huffman Encoding. [2]
- b. When is fixed length encoding preferable? [1]
- c. What is a prefix property in Huffman encoding? Will there be any ambiguities when decoding from a Huffman tree? [2]
- d. Does Huffman Encoding always reduce size? [1]

- 2a. Does Kruskal's algorithm work on directed graphs? [1]
- b. Do Minimum Spanning Tree and shortest path give the same result? Justify your reasoning using a small simulation. [2]
- c. Can Dijkstra be used to find MST? Using one line, justify your reasoning. [1]
- d. Are Kruskal's and Dijkstra algorithms both greedy algorithms? [2]

- 3. By simulating, find the Longest Common Subsequence and the length of LCS for the words: "network" and "newton". For this problem, use the upper cell's help when the characters mismatch. [4]

- 4. Suppose you are managing a delivery truck for a wholesale supply company. Your truck can carry a maximum of 9 tonnes. The goods available for transport are stored in large bulk containers and you may load any amount from a container depending on available space. [4]

The items are given in the format: Weight (Value):

3 (5), 2 (4), 6 (1), 5 (4), 1 (4)